



Topic: Logic Modeling and Normalization

Exercise 1

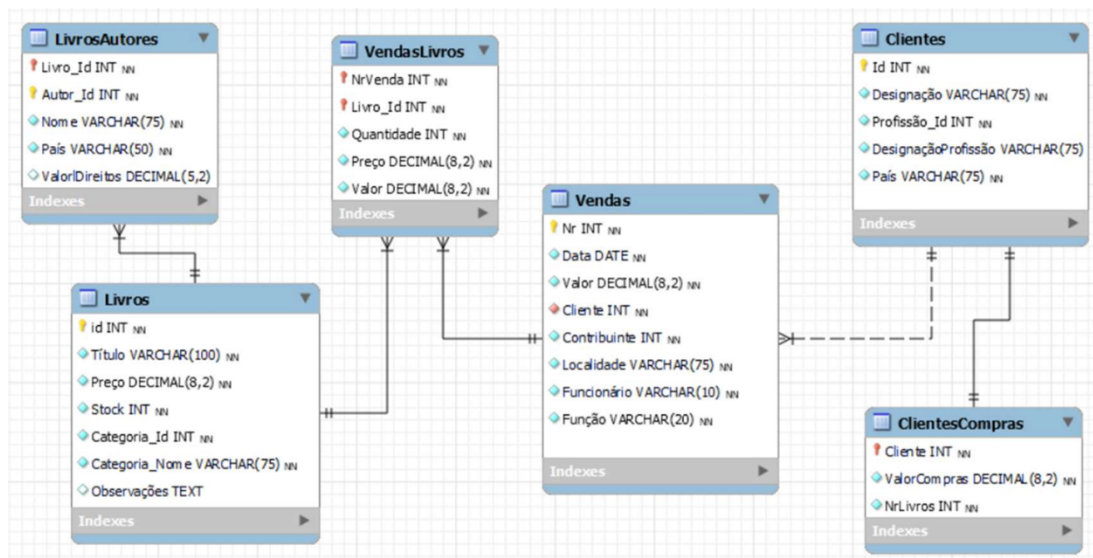


Figure 1 – Logical schema of a bookstore's database. (Belo, O.)

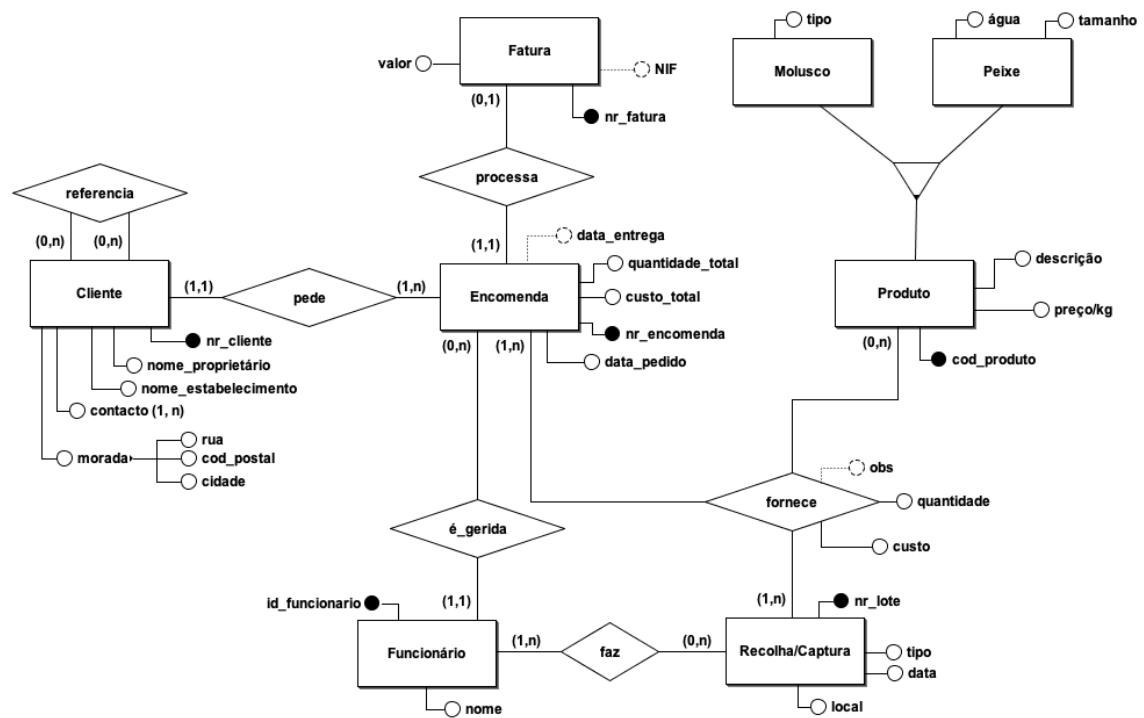
Table 1 – Functional dependencies between the various data elements of each table. (Belo, O.)

Livros	$Id \rightarrow \text{Título, Preço, Stock, Categoria_Id, Observações}$ $\text{Categoria_Id} \rightarrow \text{Categoria_Nome}$
LivrosAutores	$\text{Livro_Id, Autor_Id} \rightarrow \text{Livro_Id, Autor_id}$ $\text{Livro_Id, Autor_Id} \rightarrow \text{ValorDireitos}$ $\text{Autor_Id} \rightarrow \text{Nome, País}$
Vendas	$Nr \rightarrow \text{Data, Valor, Cliente, Funcionário}$ $\text{Cliente} \rightarrow \text{Contribuinte, Localidade}$ $\text{Funcionário} \rightarrow \text{Função}$
VendasLivros	$\text{NrVenda, Livro_Id} \rightarrow \text{Quantidade, Preço, Valor}$
Clientes	$Id \rightarrow \text{Designação, Profissão, País}$ $\text{Profissão} \rightarrow \text{DesignaçãoProfissão}$
ClientesCompras	$\text{Cliente} \rightarrow \text{ValorCompras, NrLivros}$

Consider the logical scheme presented in Figure 1, relating to a database on book sales that were made in a bookstore, and the list of functional dependencies presented in Table 1. Carefully and thoroughly analyze the schema presented, as well as the functional dependencies listed. Then, do the following exercises:

- Demonstrate that the scheme presented is or is not standardized. Justify.
- If the system is not normalized by 3FN (third normal form), normalize it. Justify the steps you take throughout the standardization process.

Exercise 2



Convert the presented conceptual schema to a possible logical schema. List and justify each of the steps taken in the conversion process.